



GOA COLLEGE OF ENGINEERING

"BHAUSAHEB BANDODKAR TECHNICAL EDUCATION COMPLEX"
FARMAGUDI, PONDA- GOA - INDIA

MATHEMATICS-III UNIT 1: MATRICES LECTURE 6

DEVELOPED BY MATHEMATICS FACULTY
DEPARTMENT OF SCIENCE & HUMANITIES
GOA COLLEGE OF ENGINEERING



MATRICES LECTURE 6

TOPICS TO BE COVERED:

- Eigenvalues and Eigenvectors of the matrix.
- Properties of Eigenvalues and Eigenvectors.
- Cayley-Hamilton Theorem
- Application of Cayley-Hamilton Theorem.

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Eigenvalues and Eigenvectors

Characteristic Matrix:

For a given square matrix A of order n ,

$A - \lambda I$ matrix is a matrix of order n called the *characteristic matrix*,

Where λ is scalar and I is the unit matrix of order n (Identity matrix).

$$\text{Ex. Let } A = \begin{bmatrix} 2 & 2 & 1 \\ 1 & 3 & 1 \\ 1 & 2 & 2 \end{bmatrix}$$

$$\text{Then } A - \lambda I = \begin{bmatrix} 2 & 2 & 1 \\ 1 & 3 & 1 \\ 1 & 2 & 2 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 2 - \lambda & 2 & 1 \\ 1 & 3 - \lambda & 1 \\ 1 & 2 & 2 - \lambda \end{bmatrix}$$

Is the Characteristic Matrix.



Eigenvalues and Eigenvectors

Characteristic Polynomial:

The determinant of characteristic matrix is nothing but Characteristic Polynomial.
i.e., The determinant $|A - \lambda I|$ when expanded will give a polynomial of degree n , which we call as characteristic polynomial of matrix A .

Ex. Let $A = \begin{bmatrix} 2 & 2 & 1 \\ 1 & 3 & 1 \\ 1 & 2 & 2 \end{bmatrix}$ then $A - \lambda I = \begin{bmatrix} 2 - \lambda & 2 & 1 \\ 1 & 3 - \lambda & 1 \\ 1 & 2 & 2 - \lambda \end{bmatrix}$ is the

Characteristic Matrix,

$$\begin{aligned} \text{Then } |A - \lambda I| &= \begin{vmatrix} 2 - \lambda & 2 & 1 \\ 1 & 3 - \lambda & 1 \\ 1 & 2 & 2 - \lambda \end{vmatrix} \\ &= (2 - \lambda)[(3 - \lambda)(2 - \lambda) - 2] - 2[(2 - \lambda) - 1] + 1[2 - (3 - \lambda)] \\ &= \lambda^3 - 7\lambda^2 + 11\lambda - 5 \text{ is the Characteristic Polynomial.} \end{aligned}$$



Eigenvalues and Eigenvectors

Characteristic Equation:

The **equation** $|A - \lambda I| = 0$ is called the Characteristic equation of the matrix A .

Ex. Let $A = \begin{bmatrix} 2 & 2 & 1 \\ 1 & 3 & 1 \\ 1 & 2 & 2 \end{bmatrix}$ then $A - \lambda I = \begin{bmatrix} 2 - \lambda & 2 & 1 \\ 1 & 3 - \lambda & 1 \\ 1 & 2 & 2 - \lambda \end{bmatrix}$ is the
Characteristic Matrix,

$$|A - \lambda I| = \begin{vmatrix} 2 - \lambda & 2 & 1 \\ 1 & 3 - \lambda & 1 \\ 1 & 2 & 2 - \lambda \end{vmatrix} = \lambda^3 - 7\lambda^2 + 11\lambda - 5 \text{ is the Characteristic Polynomial.}$$

Then $\lambda^3 - 7\lambda^2 + 11\lambda - 5 = 0$ is the **Characteristic Equation**.



Eigenvalues and Eigenvectors

Characteristic Roots or Eigenvalues:

The roots of characteristic equation $|A - \lambda I| = 0$ are called Characteristic Roots or Eigenvalues of matrix.

Ex. Let $A = \begin{bmatrix} 2 & 2 & 1 \\ 1 & 3 & 1 \\ 1 & 2 & 2 \end{bmatrix}$ then $\lambda^3 - 7\lambda^2 + 11\lambda - 5 = 0$ is the Characteristic Equation of A and

$$\lambda^3 - 7\lambda^2 + 11\lambda - 5 = 0$$

$$(\lambda - 1)(\lambda - 1)(\lambda - 5) = 0$$

$\therefore \lambda = 1, 1, 5$ and these values of λ 's are called **Characteristic Roots or Eigenvalues** of A.



Eigenvalues and Eigenvectors

Eigenvectors:

Let A be a square matrix of order n .

Let $\lambda_1, \lambda_2 \dots \lambda_n$ be n eigenvalues of the matrix A .

If there exist a non – zero column matrix X (i.e., matrix of order $n \times 1$) corresponding to eigenvalue λ such that

$$(A - \lambda I)X = 0 \quad \text{or} \quad AX = \lambda X$$

Then that non-zero column matrix X is called **eigenvector corresponding to eigenvalue λ** .

Therefore for every square matrix of order n there exist n Eigenvectors $X_1, X_2, \dots X_n$ corresponding to the Eigenvalues $\lambda_1, \lambda_2 \dots, \lambda_n$.



Examples

1. Find the Eigenvalues and Eigenvectors of $A = \begin{bmatrix} 1 & 1 & 3 \\ 1 & 5 & 1 \\ 3 & 1 & 1 \end{bmatrix}$.

Solution : The Characteristic equation of A is $|A - \lambda I| = 0$.

$$\begin{vmatrix} 1 - \lambda & 1 & 3 \\ 1 & 5 - \lambda & 1 \\ 3 & 1 & 1 - \lambda \end{vmatrix} = 0$$

$$(1 - \lambda)[(5 - \lambda)(1 - \lambda) - 1] - 1[(1 - \lambda) - 3] + 3[1 - 3(5 - \lambda)] = 0$$

$$\lambda^3 - 7\lambda^2 + 36 = (\lambda + 2)(\lambda - 3)(\lambda - 6) = 0$$

$\lambda = -2, 3, 6$ are the roots of Characteristic Equations.

$\lambda_1 = -2, \lambda_2 = 3$ and $\lambda_3 = 6$ are the **Eigenvalues** of A.



To find Eigenvectors:

Consider $(A - \lambda I)X = 0$

$$\begin{bmatrix} 1 - \lambda & 1 & 3 \\ 1 & 5 - \lambda & 1 \\ 3 & 1 & 1 - \lambda \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \text{ ——— (1)}$$

i) For X_1 put $\lambda = -2$ in equation (1)

$$\begin{bmatrix} 3 & 1 & 3 \\ 1 & 7 & 1 \\ 3 & 1 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$3x_1 + x_2 + 3x_3 = 0$$

$$x_1 + 7x_2 + x_3 = 0$$

$$3x_1 + x_2 + 3x_3 = 0$$

Solve by using Cross-Multiplication Method

$$\frac{x_1}{\begin{vmatrix} 1 & 3 \\ 7 & 1 \end{vmatrix}} = \frac{x_2}{-\begin{vmatrix} 3 & 3 \\ 1 & 1 \end{vmatrix}} = \frac{x_3}{\begin{vmatrix} 3 & 1 \\ 1 & 7 \end{vmatrix}}$$

Cross Multiplication method:

Consider

$$a_1x_1 + b_1x_2 + c_1x_3 = 0$$

$$a_2x_1 + b_2x_2 + c_2x_3 = 0$$

Two equations which are not equal or not multiple of each other.

$$\text{Then } \frac{x_1}{\begin{vmatrix} b_1 & c_1 \\ b_2 & c_2 \end{vmatrix}} = \frac{x_2}{-\begin{vmatrix} a_1 & c_1 \\ a_2 & c_2 \end{vmatrix}} = \frac{x_3}{\begin{vmatrix} a_1 & b_1 \\ a_2 & b_2 \end{vmatrix}}$$



$$\frac{x_1}{-20} = \frac{x_2}{0} = \frac{x_3}{20} \Rightarrow \frac{x_1}{-1} = \frac{x_2}{0} = \frac{x_3}{1}$$

Therefore the Eigenvector corresponding to eigenvalue $\lambda_1 = -2$ is $X_1 = \begin{bmatrix} -k \\ 0 \\ k \end{bmatrix}$ or $\begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}$.

ii) For X_2 put $\lambda = 3$ in equation (1)

$$\begin{bmatrix} -2 & 1 & 3 \\ 1 & 2 & 1 \\ 3 & 1 & -2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$-2x_1 + x_2 + 3x_3 = 0$$

$$x_1 + 2x_2 + x_3 = 0$$

$$3x_1 + x_2 - 2x_3 = 0$$

From first and second equation,

$$\frac{x_1}{\begin{vmatrix} 1 & 3 \\ 2 & 1 \end{vmatrix}} = \frac{x_2}{-\begin{vmatrix} -2 & 3 \\ 1 & 1 \end{vmatrix}} = \frac{x_3}{\begin{vmatrix} -2 & 1 \\ 1 & 2 \end{vmatrix}} \Rightarrow \frac{x_1}{-5} = \frac{x_2}{5} = \frac{x_3}{-5} \Rightarrow \frac{x_1}{-1} = \frac{x_2}{1} = \frac{x_3}{-1}$$

Therefore the Eigenvector corresponding to eigenvalue $\lambda_2 = 3$ is $X_2 = \begin{bmatrix} -k \\ k \\ -k \end{bmatrix}$ or $\begin{bmatrix} -1 \\ 1 \\ -1 \end{bmatrix}$



ii) For X_2 put $\lambda = 6$ in equation (1)

$$\begin{bmatrix} -5 & 1 & 3 \\ 1 & -1 & 1 \\ 3 & 1 & -5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$-5x_1 + x_2 + 3x_3 = 0$$

$$x_1 - x_2 + x_3 = 0$$

$$3x_1 + x_2 - 5x_3 = 0$$

From first and second equation,

$$\frac{x_1}{\begin{vmatrix} 1 & 3 \\ -1 & 1 \end{vmatrix}} = \frac{x_2}{-\begin{vmatrix} -5 & 3 \\ 1 & 1 \end{vmatrix}} = \frac{x_3}{\begin{vmatrix} -5 & 1 \\ 1 & -1 \end{vmatrix}} \Rightarrow \frac{x_1}{4} = \frac{x_2}{8} = \frac{x_3}{4} \Rightarrow \frac{x_1}{1} = \frac{x_2}{2} = \frac{x_3}{1}$$

Therefore the Eigenvector corresponding to eigenvalue $\lambda_3 = 6$ is $X_3 = \begin{bmatrix} k \\ 2k \\ k \end{bmatrix}$ or $\begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}$



Properties of Eigenvalues and Eigenvectors

- The sum of the Eigenvalue is equal to the sum of its diagonal elements i.e trace of the matrix.
- The product of the eigenvalues is nothing but determinant of the matrix.
- The square matrix A is singular iff zero is an eigenvalue of A .
- A square matrix and its transpose have the same eigenvalues.
- If λ is an Eigenvalue of a square matrix A then $\lambda + k$ is eigenvalue of $A + kI$.
- Let $\lambda_1, \lambda_2, \dots, \lambda_n$ be Eigenvalues of square matrix A of order n . Then
 1. $c\lambda_1, c\lambda_2, \dots, c\lambda_n$ are Eigenvalues of (cA) .
 2. $\lambda_1^m, \lambda_2^m, \dots, \lambda_n^m$ are Eigenvalues of A^m .
 3. $\frac{1}{\lambda_1}, \frac{1}{\lambda_2}, \dots, \frac{1}{\lambda_n}$ are eigenvalues of A^{-1} .
 4. $\frac{|A|}{\lambda_1}, \frac{|A|}{\lambda_2}, \dots, \frac{|A|}{\lambda_n}$ are eigenvalues of $\text{Adj } A$.



Properties of Eigenvalues and Eigenvectors

- Corresponding to an Eigenvalue λ of a square matrix A there exists more than one Eigenvector.
- The Eigenvalues of Hermitian matrix are real number.
- The Eigenvalues of Symmetric matrix are real number.
- The Eigenvalues of Skew-symmetric matrix are either zero or purely imaginary.
- The Eigenvalues of an Orthogonal matrix are ± 1 .
- Corresponding to an Eigenvector X of a square matrix there exist one and only one Eigenvalue.
- Eigenvectors corresponding to different eigenvalues are linearly independent.



Properties of Eigenvalues and Eigenvectors

- If all the eigenvalues $\lambda_1, \lambda_2, \dots, \lambda_n$ of a matrix A are distinct, then the corresponding eigenvectors $X_1, X_2, \dots, X_n$ are linearly independent.
- If two or more eigenvalues are equal, then corresponding eigenvectors may be linearly independent or dependent.

Orthogonal Vectors: Two column vectors $X = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$ and $Y = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix}$ are said to be orthogonal

If $X^T Y = 0$ i.e. $x_1 y_1 + x_2 y_2 + \dots + x_n y_n = 0$

Ex. $X = \begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix}$, $Y = \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix}$ then X and Y are orthogonal.

- The eigenvectors corresponding to distinct eigenvalues of real symmetric matrix are orthogonal.



Polynomial with matrix coefficients

$$\text{Let } A = \begin{bmatrix} 1 & 1 & 3 \\ 1 & 5 & 1 \\ 3 & 1 & 1 \end{bmatrix} \text{ then } A - \lambda I = \begin{bmatrix} 1 - \lambda & 1 & 3 \\ 1 & 5 - \lambda & 1 \\ 3 & 1 & 1 - \lambda \end{bmatrix}$$

$$\text{adj}(A - \lambda I) = \begin{bmatrix} \begin{vmatrix} 5 - \lambda & 1 \\ 1 & 1 - \lambda \end{vmatrix} & -\begin{vmatrix} 1 & 3 \\ 1 & 1 - \lambda \end{vmatrix} & \begin{vmatrix} 1 & 3 \\ 5 - \lambda & 1 \end{vmatrix} \\ -\begin{vmatrix} 1 & 1 \\ 3 & 1 - \lambda \end{vmatrix} & \begin{vmatrix} 1 - \lambda & 3 \\ 3 & 1 - \lambda \end{vmatrix} & -\begin{vmatrix} 1 - \lambda & 3 \\ 1 & 1 \end{vmatrix} \\ \begin{vmatrix} 1 & 5 - \lambda \\ 3 & 1 \end{vmatrix} & -\begin{vmatrix} 1 - \lambda & 1 \\ 3 & 1 \end{vmatrix} & \begin{vmatrix} 1 - \lambda & 1 \\ 1 & 5 - \lambda \end{vmatrix} \end{bmatrix}$$

$$= \begin{bmatrix} \lambda^2 - 6\lambda - 4 & \lambda + 2 & 3\lambda - 14 \\ \lambda + 2 & \lambda^2 - 2\lambda - 8 & \lambda + 2 \\ 3\lambda - 14 & \lambda + 2 & \lambda^2 - 6\lambda - 4 \end{bmatrix} = \lambda^2 \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} + \lambda \begin{bmatrix} -6 & 1 & 3 \\ 1 & -2 & 1 \\ 3 & 1 & -6 \end{bmatrix} + \begin{bmatrix} -4 & 2 & -14 \\ 2 & -8 & 2 \\ -14 & 2 & -4 \end{bmatrix}$$

$= \lambda^2 A + \lambda B + C$ where A, B and C are square matrices of order 3x3.



Cayley-Hamilton Theorem

Statement: Every square matrix satisfies its own Characteristic equation.

Or If A is square matrix of order n , and $|A - \lambda I| = 0$ or

$(-1)^n \lambda^n + k_1 \lambda^{n-1} + \dots + k_{n-1} \lambda + k_n = 0$ is *Characteristic equation*

Then $(-1)^n A^n + k_1 A^{n-1} + \dots + k_{n-1} A + k_n I = 0$

Proof: Consider the matrix $(A - \lambda I)$ and $P = \text{adj}(A - \lambda I)$.

Since the elements of $A - \lambda I$ are at most of the first degree in λ , the elements of $\text{adj}(A - \lambda I)$ are at most degree $(n - 1)$ in λ .

Thus, $\text{adj}(A - \lambda I)$ may be written as a matrix polynomial in λ , given by

$$\text{adj}(A - \lambda I) = P = B_0 \lambda^{n-1} + B_1 \lambda^{n-2} + \dots + B_{n-1}$$

where $B_0, B_1 \dots B_{n-1}$ are $n \times n$ matrix, their elements being polynomial in λ .



We know that $(A - \lambda I)\text{adj}(A - \lambda I) = |(A - \lambda I)|I$

$$(A - \lambda I)(B_0\lambda^{n-1} + B_1\lambda^{n-2} + \cdots + B_{n-1}) =$$

$$[(-1)^n\lambda^n + k_1\lambda^{n-1} + \cdots + k_{n-1}\lambda + k_n].I$$

Equating Coefficient of like powers of λ on both the sides we get

$$-IB_0 = (-1)^n.I$$

$$AB_0 - IB_1 = k_1.I$$

$$AB_1 - IB_2 = k_2.I$$

.....

$$AB_{n-1} = k_n.I$$

On multiplying the equations by

$A^n, A^{n-1}, \dots, I$ respectively and adding, we obtain

$$0 = (-1)^n A^n + k_1 A^{n-1} + \cdots + k_{n-1} A + k_n I$$

Thus $(-1)^n A^n + k_1 A^{n-1} + \cdots + k_{n-1} A + k_n I = 0$.



Application

- To find A^{-1} :

By Cayley Hamilton Theorem,

$$(-1)^n A^n + k_1 A^{n-1} + \dots + k_{n-1} A + k_n I = \mathbf{0} .$$

Multiply this equation by A^{-1} ,

$$[(-1)^n A^n + k_1 A^{n-1} + \dots + k_{n-1} A + k_n I] \cdot A^{-1} = \mathbf{0} .$$

$$k_n I \cdot A^{-1} = -[(-1)^n A^{n-1} + k_1 A^{n-2} + \dots + k_{n-1} I]$$

$$A^{-1} = \frac{-1}{k_n} [(-1)^n A^{n-1} + k_1 A^{n-2} + \dots + k_{n-1} I]$$



Examples

Verify Cayley hamilton Theorem for the matrix

$$A = \begin{bmatrix} 2 & -1 & 1 \\ -1 & 2 & -1 \\ 1 & -1 & 2 \end{bmatrix} \text{ and hence find } A^{-1}.$$

Solution: Characteristic equation is $|A - \lambda I| = 0$

$$\begin{vmatrix} 2 - \lambda & -1 & 1 \\ -1 & 2 - \lambda & -1 \\ 1 & -1 & 2 - \lambda \end{vmatrix} = 0$$

$$(2 - \lambda)[(2 - \lambda)(2 - \lambda) - 1] + 1[(-1)(2 - \lambda) + 1] + 1[1 - (2 - \lambda)] = 0$$
$$-\lambda^3 + 6\lambda^2 - 9\lambda + 4 = 0$$

$$\text{Or } \lambda^3 - 6\lambda^2 + 9\lambda - 4 = 0$$



By Cayley Hamilton Theorem $A^3 - 6A^2 + 9A - 4I = 0$

Verification:

$$A^2 = \begin{bmatrix} 2 & -1 & 1 \\ -1 & 2 & -1 \\ 1 & -1 & 2 \end{bmatrix} \begin{bmatrix} 2 & -1 & 1 \\ -1 & 2 & -1 \\ 1 & -1 & 2 \end{bmatrix} = \begin{bmatrix} 6 & -5 & 5 \\ -5 & 6 & -5 \\ 5 & -5 & 6 \end{bmatrix}$$

$$A^3 = \begin{bmatrix} 2 & -1 & 1 \\ -1 & 2 & -1 \\ 1 & -1 & 2 \end{bmatrix} \begin{bmatrix} 6 & -5 & 5 \\ -5 & 6 & -5 \\ 5 & -5 & 6 \end{bmatrix} = \begin{bmatrix} 22 & -21 & 21 \\ -21 & 22 & -21 \\ 21 & -21 & 22 \end{bmatrix}$$

$$A^3 - 6A^2 + 9A - 4I =$$

$$= \begin{bmatrix} 22 & -21 & 21 \\ -21 & 22 & -21 \\ 21 & -21 & 22 \end{bmatrix} - 6 \begin{bmatrix} 6 & -5 & 5 \\ -5 & 6 & -5 \\ 5 & -5 & 6 \end{bmatrix} + 9 \begin{bmatrix} 2 & -1 & 1 \\ -1 & 2 & -1 \\ 1 & -1 & 2 \end{bmatrix} - 4 \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 22 - 36 + 18 - 4 & -21 + 30 - 9 - 0 & 21 - 30 + 9 - 0 \\ -21 + 30 - 9 - 0 & 22 - 36 + 18 - 4 & -21 + 30 - 9 - 0 \\ 21 - 30 + 9 - 0 & -21 + 30 - 9 - 0 & 22 - 36 + 18 - 4 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$



Inverse of a matrix:

$$A^3 - 6A^2 + 9A - 4I = 0$$

On multiplying by A^{-1} , we get

$$A^2 - 6A + 9I - 4IA^{-1} = 0$$

$$4A^{-1} = A^2 - 6A + 9I$$

$$4A^{-1} = \begin{bmatrix} 6 & -5 & 5 \\ -5 & 6 & -5 \\ 5 & -5 & 6 \end{bmatrix} - 6 \begin{bmatrix} 2 & -1 & 1 \\ -1 & 2 & -1 \\ 1 & -1 & 2 \end{bmatrix} + 9 \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$A^{-1} = \frac{1}{4} \begin{bmatrix} 3 & 1 & -1 \\ 1 & 3 & 1 \\ -1 & 1 & 3 \end{bmatrix}$$



Practice Problems

1. Find Eigenvalues and Eigenvectors of the following matrix:

$$\begin{bmatrix} 1 & 0 & -1 \\ 1 & 2 & 1 \\ 2 & 2 & 3 \end{bmatrix}, \begin{bmatrix} -2 & 2 & -3 \\ 2 & 1 & -6 \\ -1 & -2 & 0 \end{bmatrix}, \begin{bmatrix} 3 & -1 & 1 \\ -1 & 5 & -1 \\ 1 & -1 & 3 \end{bmatrix}, \begin{bmatrix} 1 & 1 & 1 \\ 0 & 2 & 1 \\ -4 & 4 & 3 \end{bmatrix}$$

2. Verify Cayley Hamilton for each of the following matrix and hence find A^{-1} and A^{-2} :

$$\begin{bmatrix} 1 & 1 & 3 \\ 1 & 3 & -3 \\ -2 & -4 & -4 \end{bmatrix}, \begin{bmatrix} 1 & 0 & 3 \\ 2 & 1 & -1 \\ 1 & -1 & 1 \end{bmatrix}, \begin{bmatrix} 2 & -1 & 1 \\ -1 & 2 & -1 \\ 1 & -1 & 2 \end{bmatrix}, \begin{bmatrix} 1 & 1 & 2 \\ 3 & 1 & 1 \\ 2 & 3 & 1 \end{bmatrix}$$

Thank You